Kripal krishna

Mentor: Vishal Gupta

Capstone Project

E-Cart Automation Testing Project

Team name :

Automation Warriors

Git hub: https://github.com/kpranaysimha2004-tech/Pixel-cart

Meet Our Team

Automation Warriors • Pixel-cart

Kripal Krishna Sri Prasad Behera
Test Automation Lead

Background: Computer Science & automation testing

Passionate about: Selenium, Python, and QA frameworks

Sohan Paira
Automation Engineer

Background: Web testing and authentication workflows

Passionate about: Functional testing and bug finding

Arnab Sarkar
QA Engineer

Background: BDD scenarios and user-flow validation

Passionate about: Test design and quality assurance

Kotupalli Pranaya Simha
Test Automation Engineer

Background: Product and UI automation testing

Passionate about: Scalable automation and clean code

Deepak Ghoti
Automation Engineer

Background: Cart workflows and feature validation

Passionate about: Reliable testing and broad coverage

Jaideep Mandal
QA Engineer

Background: Checkout and validation testing

Passionate about: Solving real-world QA challenges

Vanshika Rawat
Test Engineer

Background: Payment flow and form validation

Passionate about: Detail-oriented testing and UX quality

Kodi Tulasi
QA Engineer

Background: UI verification and test reporting

Passionate about: Reporting, documentation, and quality delivery

Together, We Ensure Quality!

A passionate team committed to delivering reliable automation solutions and high-quality software.

Team Contribution

Module Ownership – Pixel-cart

Kripal Krishna Sri Prasad Behera

- Framework setup
- Project structure
- conftest.py & dependencies
- Final report coordination

Sohan Paira

- Authentication module
- Login scenarios
- Invalid login validation
- Empty credentials validation

Arnab Sarkar

- Signup & logout module
- User registration
- Dynamic user handling
- Logout verification

Kotupalli Pranaya Simha

- Product browsing module
- Product categories
- Product details
- Product navigation

Deepak Ghoti

- Cart module
- Add / remove product
- Cart product verification
- Empty cart validation

Jaideep Mandal

- Cart total & checkout
- Multiple products
- Total price calculation
- Place order flow

Vanshika Rawat

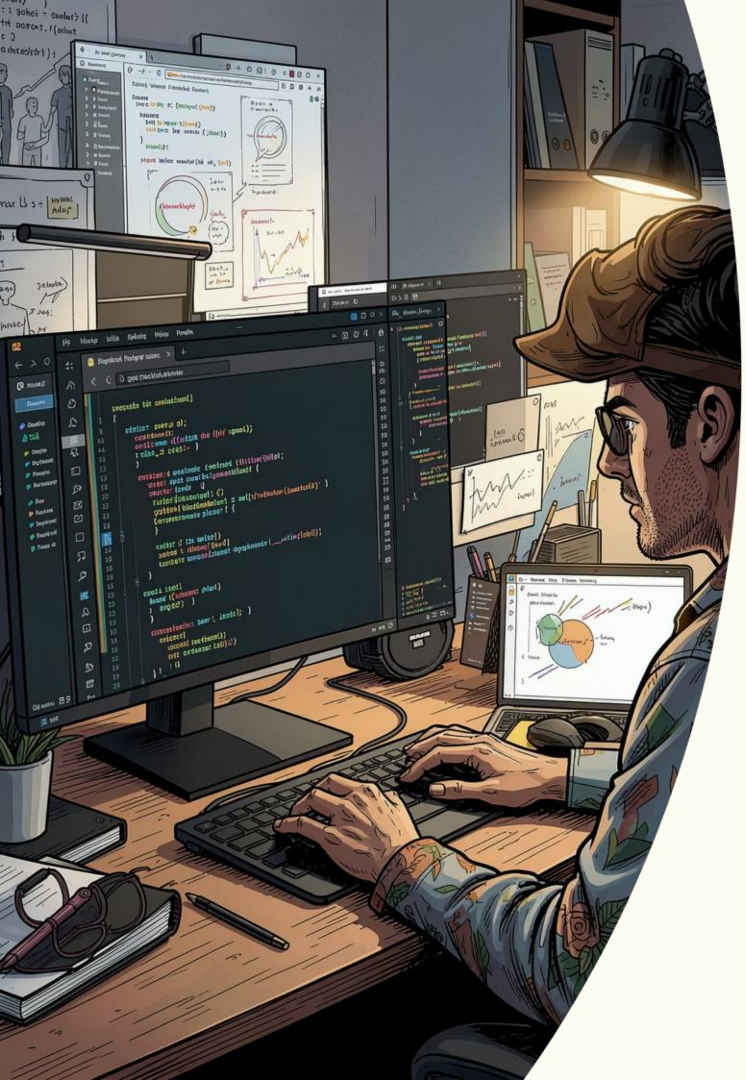
- Payment & form validation
- Valid payment details
- Empty order validation
- Cancel payment form

Kodi Tulasi

- UI validation & reporting
- About Us popup
- Contact form & carousel
- HTML report verification

Pixel-cart

E-commerce Automation Testing Project — Built with Python, Selenium, Pytest-BDD & Page Object Model



Project Introduction

Pixel-cart is an e-commerce automation testing project developed using **Python, Selenium WebDriver, Pytest-BDD, and Page Object Model**. It automates important user flows like login, signup, product browsing, cart operations, checkout, and payment validation.

Key Highlights

Selenium with Python
Built using Selenium with Python.

BDD & POM
Follows BDD and Page Object Model.

E-commerce Workflows
Covers major e-commerce workflows.

HTML Reports
Generates HTML test execution report.

Problem Statement

E-commerce applications have many important workflows such as user login, product search, cart management, and checkout. Testing these features manually again and again takes more time and may lead to mistakes.

Challenges in Manual Testing

- Repeated testing of the same features is time-consuming.
- Human errors can occur during validation.
- It is difficult to test multiple scenarios quickly.
- Regression testing becomes slow when the application grows.
- Bugs in cart, checkout, or payment flow can affect user experience.

Need for Automation

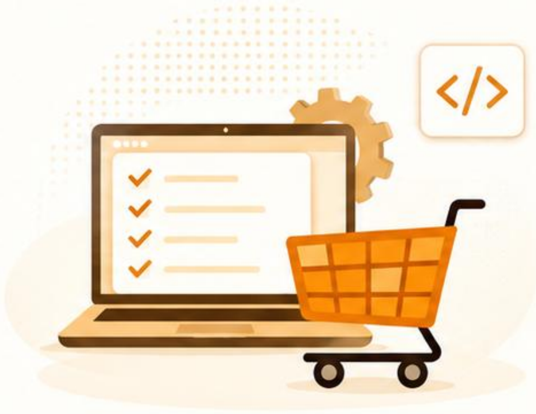
To solve these problems, we created an automation testing framework that can execute test scenarios faster, improve accuracy, and generate proper reports for review.

Objectives of Pixel-cart Automation

The main objective of Pixel-cart is to automate the core functionalities of an e-commerce application and validate them using a structured testing framework.

Project Objectives

- Automate important e-commerce workflows.
- Validate login, signup, logout, product, cart, and checkout features.
- Use BDD scenarios to make test cases more readable.
- Apply Page Object Model to separate locators and test logic.
- Use explicit waits to improve stability.
- Generate HTML reports for test execution results.
- Reduce manual testing effort and improve test reliability.



Final Goal: To build a maintainable automation framework that gives quick feedback about application quality.

Technology Stack Used

Pixel-cart is developed using modern automation testing tools and frameworks.

Tools and Technologies

Category	Technology Used
Programming Language	Python
Automation Tool	Selenium WebDriver
Testing Framework	Pytest
BDD Framework	Pytest-BDD
Design Pattern	Page Object Model
Reporting	Pytest-HTML / HTML Report
Browser	Google Chrome
Version Control	Git and GitHub
Test Format	Gherkin Feature Files

Framework Architecture

The Pixel-cart framework is organized in a clean folder structure so that test cases, page classes, and step definitions are separated properly.

Repository Structure

```
Pixel-cart/
├── assets/
├── features/
├── pages/
├── steps/
├── conftest.py
├── requirements.txt
└── .gitignore
```

Folder Explanation

- **features/** — Contains BDD scenarios written in Gherkin language.
- **pages/** — Contains Page Object Model classes for different application pages.
- **steps/** — Contains step definition files that connect feature steps with Python code.
- **assets/** — Contains test reports and report styling files.
- **conftest.py** — Contains browser setup, WebDriver fixture, wait helper, and alert handling.
- **requirements.txt** — Contains all required Python dependencies for the project.

Architecture Benefit: This structure makes the framework easy to understand, easy to maintain, and suitable for teamwork.

Functionalities Automated

Pixel-cart covers the major functional areas of an e-commerce application.

Authentication

- User signup
- Valid login
- Invalid login
- Empty credential validation
- Logout validation

Product Browsing

- View phone, laptop, and monitor products
- Open product details page
- Validate product name, price, and description

Cart Operations

- Add and remove products from cart
- Verify product in cart
- Add multiple products
- Validate empty cart and total price

Order & Payment

- Place order with valid details
- Validate empty order form
- Cancel payment form
- Verify order success popup

UI & Content Validation

- About Us popup
- Contact form submission
- Homepage carousel navigation

Test Scenarios Covered

The project includes BDD test scenarios for the main e-commerce workflows.

Major Test Scenarios

- Login, signup, and logout validation.
- Product category and product detail verification.
- Add, remove, and verify products in cart.
- Cart total price calculation.
- Checkout and payment validation.
- Contact form, About Us popup, and carousel testing.

BDD Advantage

The scenarios are written using simple **Given, When, Then** steps, making them easy to understand and present.

Test Execution & Outcomes

After completing the automation scripts, we executed the full test suite and generated an HTML report.

Execution Summary

32	32	0	0	100%
Total Test Cases	Passed Test Cases	Failed Test Cases	Skipped Test Cases	Pass Rate

Outcome

- All major workflows were successfully automated.
- The test suite executed without failures.
- HTML report was generated for result analysis.
- Framework is reusable and easy to extend.
- Regression testing can be performed faster.

Final Result

Pixel-cart achieved **32/32** passed test cases with a **100%** pass rate.

Test Report

TEST EXECUTION SUMMARY

Passed Testcases Report

This report summarizes executed test cases and highlights passed results from the JUnit output, delivering a clean status overview for stakeholders.

32
Test cases executed

32
Total test cases in suite

32
Passed tests included

100%
Pass rate

0
Failed tests

0
Skipped tests

Passed Skipped Failed/Error

JUnit source: `results.xml` · Generated: 2026-05-20T11:18:24

Git hub Repository link : https://github.com/kpranaysimha2004-tech/Pixel-cart

Conclusion

Pixel-cart is a complete e-commerce automation testing project built using Selenium with Python and Pytest -BDD. The project follows Page Object Model, which makes the code reusable, organized , and easy to maintain.

Through this project, we automated important workflows like authentication, product browsing, cart operations, checkout, payment, and UI validations.

The final execution result shows that all **32 test cases passed successfully**, proving that the framework is stable and effective for regression testing.

THANK YOU!

FOR YOUR TIME AND ATTENTION

? OPEN FOR
QUESTIONS AND FEEDBACK.

For queries : k.pranaysimha2004@gmail.com